

Library-Management-System

Database connectivity (My SQL WorkBench):

1.Members List

The screenshot shows the MySQL Workbench interface with the 'librarymanagement_db' schema selected in the Navigator. The SQL editor contains the query `select * from MEMBERS;`. The Result Grid displays the following data:

member_id	name	email	books_borrowed
101	Alice Johnson	alice.johnson@example.com	1
102	Bob Smith	bob.smith@example.com	2
103	Charlie Brown	charlie.brown@example.com	3
104	Diana Prince	diana.prince@example.com	4
105	Ethan Hunt	ethan.hunt@example.com	5
106	Fiona Glenanne	fiona.g@example.com	6
107	George Martin	george.martin@example.com	7
108	Hannah Wells	hannah.wells@example.com	8
109	Ivan Drago	ivan.drago@example.com	9
110	Julia Meade	julia.meade@example.com	10
NULL	NULL	NULL	NULL

2.Books List

The screenshot shows the MySQL Workbench interface with the 'librarymanagement_db' schema selected in the Navigator. The SQL editor contains the query `select * from BOOKS;`. The Result Grid displays the following data:

book_id	title	author	price
1	Java Programming	James Gosling	499.99
2	Effective Java	Joshua Bloch	599.5
3	Clean Code	Robert C. Martin	699
4	Head First Java	Kathy Sierra	450
5	Data Structures	Seymour Lipschutz	399.75
6	Operating Systems	Abraham Silberschatz	520
7	Database Systems	Raghu Ramakrishnan	580.2
8	Computer Networks	Andrew S. Tanenbaum	630.45
9	C Programming	Dennis Ritchie	350
10	travel	arjo	778
11	Artificial Intelligence	Stuart Russell	675.8
NULL	NULL	NULL	NULL

Books List (Console)

```
Problems Javadoc Declaration Console X
Main (1) [Java Application] C:\Users\Mitodru\Downloads\eclipse-java-2025-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.21.0.7.v20250502-0916\jre\bin\javaw.exe (29-Jul-2025)

===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 3

--- Books ---
1 | Java Programming | James Gosling | ₹499.99
2 | Effective Java | Joshua Bloch | ₹599.5
3 | Clean Code | Robert C. Martin | ₹699.0
4 | Head First Java | Kathy Sierra | ₹450.0
5 | Data Structures | Seymour Lipschutz | ₹399.75
6 | Operating Systems | Abraham Silberschatz | ₹520.0
7 | Database Systems | Raghu Ramakrishnan | ₹580.2
8 | Computer Networks | Andrew S. Tanenbaum | ₹630.45
9 | C Programming | Dennis Ritchie | ₹350.0
10 | travel | arjo | ₹778.0
11 | Artificial Intelligence | Stuart Russell | ₹675.8
```

Members List (Console)

```
Main (1) [Java Application] C:\Users\Mitodru\Downloads\eclipse-java-2025-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.21.0.7.v20250502-0916\jre\bin\javaw.exe (29-Jul-2025)

===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 4

--- Members ---
101 | Alice Johnson | alice.johnson@example.com | 1
102 | Bob Smith | bob.smith@example.com | 2
103 | Charlie Brown | charlie.brown@example.com | 3
104 | Diana Prince | diana.prince@example.com | 4
105 | Ethan Hunt | ethan.hunt@example.com | 5
106 | Fiona Glenanne | fiona.g@example.com | 6
107 | George Martin | george.martin@example.com | 7
108 | Hannah Wells | hannah.wells@example.com | 8
109 | Ivan Drago | ivan.drago@example.com | 9
110 | Julia Meade | julia.meade@example.com | 10
501 | arjo | arjo@gmail.com | 500
```

Insert Books(Console):

```
===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 1
Enter Book ID: 500
Enter Title: Geo
Enter Author: Sukhdeb
Enter Price: 789.22
Book inserted successfully!
```

Insert Members(Console):

```
===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 2
Enter Member ID: 501
Enter Name: arjo
Enter Email: arjo@gmail.com
Enter Book Title Borrowed: 500
Member inserted successfully!
```

Update Books(Console):

```

===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 5
Enter Book ID to update: 10
Which field do you want to update? (title/author/price): title
Enter new title: Travel with arjo
Book updated.

```

All Books after the update(console):

```

Main (1) [Java Application] C:\Users\Mitodru\Downloads\eclipse-java-2025-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full...
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 3
--- Books ---
1 | Java Programming | James Gosling | ₹499.99
2 | Effective Java | Joshua Bloch | ₹599.5
3 | Clean Code | Robert C. Martin | ₹699.0
4 | Head First Java | Kathy Sierra | ₹450.0
5 | Data Structures | Seymour Lipschutz | ₹399.75
6 | Operating Systems | Abraham Silberschatz | ₹520.0
7 | Database Systems | Raghu Ramakrishnan | ₹580.2
8 | Computer Networks | Andrew S. Tanenbaum | ₹630.45
9 | C Programming | Dennis Ritchie | ₹350.0
10 | Travel with arjo | arjo | ₹778.0
11 | Artificial Intelligence | Stuart Russell | ₹675.8
500 | Geo | Sukhdeb | ₹789.22

```

Update Members(Console):

```
===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 6
Enter Member ID to update: 501
Which field do you want to update? (name/email/books_borrowed): name
Enter new name: Mitodru
Member updated.
```

All Members after the update(console):

```
Main (1) [Java Application] C:\Users\Mitodru\Downloads\eclipse-java-2025-06-R-win32-x86_64\e
===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 4
|
--- Members ---
101 | Alice Johnson | alice.johnson@example.com | 1
102 | Bob Smith | bob.smith@example.com | 2
103 | Charlie Brown | charlie.brown@example.com | 3
104 | Diana Prince | diana.prince@example.com | 4
105 | Ethan Hunt | ethan.hunt@example.com | 5
106 | Fiona Glenanne | fiona.g@example.com | 6
107 | George Martin | george.martin@example.com | 7
108 | Hannah Wells | hannah.wells@example.com | 8
109 | Ivan Drago | ivan.drago@example.com | 9
110 | Julia Meade | julia.meade@example.com | 10
501 | Mitodru | arjo@gmail.com | 500
```

Delete Members and Books(Console):

```
===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 7
Enter Member ID to delete: 501
Member deleted.

===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 8
Enter Book ID to delete: 500
Book deleted.
```

All Books and Members after deletion(Console):

```
Main (1) [Java Application] C:\Users\Mitodru\Downloads\eclipse-java-2025-06-R-win32-x86_64\eclipse\plugins\org
BOOKS
1 | Java Programming | James Gosling | ₹499.99
2 | Effective Java | Joshua Bloch | ₹599.5
3 | Clean Code | Robert C. Martin | ₹699.0
4 | Head First Java | Kathy Sierra | ₹450.0
5 | Data Structures | Seymour Lipschutz | ₹399.75
6 | Operating Systems | Abraham Silberschatz | ₹520.0
7 | Database Systems | Raghuram Krishnan | ₹580.2
8 | Computer Networks | Andrew S. Tanenbaum | ₹630.45
9 | C Programming | Dennis Ritchie | ₹350.0
10 | Travel with arjo | arjo | ₹778.0
11 | Artificial Intelligence | Stuart Russell | ₹675.8

===== Library Management Menu =====
1. Insert Book
2. Insert Member
3. View All Books
4. View All Members
5. Update Book
6. Update Member
7. Delete Member
8. Delete Book
9. Exit
Enter your choice: 4

--- Members ---
101 | Alice Johnson | alice.johnson@example.com | 1
102 | Bob Smith | bob.smith@example.com | 2
103 | Charlie Brown | charlie.brown@example.com | 3
104 | Diana Prince | diana.prince@example.com | 4
105 | Ethan Hunt | ethan.hunt@example.com | 5
106 | Fiona Glenanne | fiona.g@example.com | 6
107 | George Martin | george.martin@example.com | 7
108 | Hannah Wells | hannah.wells@example.com | 8
109 | Ivan Drago | ivan.drago@example.com | 9
110 | Julia Meade | julia.meade@example.com | 10
```